

1. (20 marks) For $(a, b) \in \mathbb{R}^2$, consider the following quadratic form

$$Q(\mathbf{x}) = a(x_1^2 + x_3^2) + bx_2^2 + 2a(x_1x_2 + x_2x_3).$$

- (a) (4 marks) Identify the *symmetric matrix of coefficients* A associated with this quadratic form. That is, find the matrix A satisfying $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$. Then, find all the principal minors of A .

Solutions: (2 marks) The matrix of coefficients is:

$$A = \begin{pmatrix} a & a & 0 \\ a & b & a \\ 0 & a & a \end{pmatrix}$$

(2 marks) Its third order principal minor is $a^2b - 2a^3$. Its second order principal minors are $ab - a^2$ and a^2 . Its first order principal minors are a and b .

- (b) (12 marks) Define what it means for A to be *indefinite*, *positive semidefinite* and *negative semidefinite*. Then, find all parameter values for a and b such that the quadratic form $Q(\mathbf{x})$ is respectively *indefinite*, *positive semidefinite* and *negative semidefinite*?

Solutions: (2 marks) A quadratic form is indefinite if $Q(\mathbf{x}) > 0$ and $Q(\mathbf{y}) < 0$ for some $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n \setminus \mathbf{0}$. A quadratic form is positive semidefinite if $Q(\mathbf{x}) \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n \setminus \mathbf{0}$. A quadratic form is negative semidefinite if $Q(\mathbf{x}) \leq 0$ for all $\mathbf{x} \in \mathbb{R}^n \setminus \mathbf{0}$.

(4 marks) $Q(\mathbf{x})$ is positive semidefinite if all principal minors are positive which requires:

$$b \geq 2a \geq 0.$$

The condition already accounts for scenarios in which some of the parameters are equal to 0.

(4 marks) $Q(\mathbf{x})$ is negative semidefinite if all principal minors suitably alternate in sign which requires:

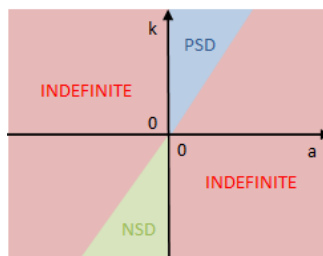
$$0 \geq 2a \geq b.$$

The condition already accounts for scenarios in which some of the parameters are equal to 0.

(2 marks) $Q(\mathbf{x})$ is indefinite whenever it is not semidefinite.

- (c) (4 marks) Plot the pairs (a, b) for which $Q(\mathbf{x})$ is respectively *indefinite*, *positive semidefinite* and *negative semidefinite* in $\mathbb{R}^2$.

Solutions: (4 marks) The plot below depicts the answer



2. (20 marks) Consider the functions $f(\mathbf{x}) = -x_1^2 x_2^2$ defined on some convex set U in $\mathbb{R}^2$.

- (a) (4 marks) Define what it means for f to be *quasi-concave* and *quasi-convex*.

Solutions: The function f is: (2 marks) *quasi-concave* on U if $C_a^+ = \{\mathbf{x} \in U \mid f(\mathbf{x}) \geq a\}$ is a convex set for every real number; (2 marks) *quasi-convex* on U if $C_a^- = \{\mathbf{x} \in U \mid f(\mathbf{x}) \leq a\}$ is a convex set for every real number a .

- (b) (16 marks) Is f *quasi-concave* or *quasi-convex* when U is equal to $\mathbb{R}^2$ and $\mathbb{R}_+^2$ respectively? To show this, you may exploit results proven in lecture.

Solutions: (8 marks) The function f is not quasi-concave on $\mathbb{R}^2$ and $\mathbb{R}_+^2$. To show this consider $\mathbf{x} = (2, 0)$ and $\mathbf{y} = (0, 2)$ and let $\mathbf{z} = \mathbf{x}/2 + \mathbf{y}/2 = (1, 1) \in \mathbb{R}_+^2$. If so, we have that

$$\mathbf{x}, \mathbf{y} \in C_0^+ \quad \text{and yet} \quad \mathbf{z} \notin C_0^+.$$

(4 marks) The function f is not quasi-convex on $\mathbb{R}^2$. To show this consider $\mathbf{x} = (1, 1)$ and $\mathbf{y} = (1, -1)$ and let $\mathbf{z} = \mathbf{x}/2 + \mathbf{y}/2 = (1, 0) \in \mathbb{R}^2$. If so, we have that

$$\mathbf{x}, \mathbf{y} \in C_{-1}^- \quad \text{and yet} \quad \mathbf{z} \notin C_{-1}^-.$$

(4 marks) The function f is quasi-convex on $\mathbb{R}_+^2$. To show this consider the increasing transformation $g(f) = -\log(-f)$. For such transformation we have that

$$h(\mathbf{x}) = g(f(\mathbf{x})) = -2 \log x_1 - 2 \log x_2.$$

The function h is convex on $\mathbb{R}_+^2$, since it is a sum of convex functions. Because h is convex and increasing transformation of h is quasi-convex on $\mathbb{R}_+^2$. Since g^{-1} is an increasing transformation, we thus have that f is quasi-convex since $f(\mathbf{x}) = g^{-1}(h(\mathbf{x}))$.

3. (40 marks) For $a > 0$, consider the problem:

$$\max_{x,y} ax + y \quad \text{s.t.} \quad x^2 + y^2 \leq 3/4, \quad x \geq 0, \quad y \geq a.$$

- (a) (7 marks) Is the objective function concave? Are the constraints convex? For what values of $a > 0$ is Slater condition satisfied? So, for what values of $a > 0$, can you solve this optimization problem using the Kuhn-Tucker method?

Solutions: (3 marks) The objective function is linear and thus concave. The first constraint is convex, since its Hessian satisfies

$$D^2 g_1(x, y) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

$$|D^2 g_1(x, y)| = 4 \geq 0, \quad 2 > 0.$$

The other constraints are linear and thus convex. (4 marks) Thus, when the constraint set has a non-empty interior, we can adopt the Kuhn-Tucker approach to solve the problem. This is the case for all $a \in (0, \sqrt{3}/2)$, since the point $\tilde{x} = \varepsilon$ and $\tilde{y} = a + \varepsilon$ is always in the interior of the constraint set for ε sufficiently small as long as $a < \sqrt{3}/2$, since

$$\tilde{x}^2 + \tilde{y}^2 = 2\varepsilon^2 + 2\varepsilon + a^2 < 3/4, \quad \tilde{x} > 0, \quad \tilde{y} > a.$$

- (b) (7 marks) Suppose the restrictions on a found in part (a) hold. Write down the Kuhn-Tucker necessary first order conditions that must be satisfied by the solution of the constrained optimization problem. Are solutions to these conditions also maximizers of the Lagrangian?

Solutions: (5 marks) The necessary first order conditions for the solution of the constrained maximization problem are:

$$\begin{aligned} 0 &= a - 2\lambda_1 x + \lambda_2, \\ 0 &= 1 - 2\lambda_1 y + \lambda_3, \\ 0 &= \lambda_1(x^2 + y^2 - 3/4) = \lambda_2 x = \lambda_3(y - a), \\ 0 &\geq x^2 + y^2 - 3/4, \quad 0 \geq -x, \quad 0 \geq a - y, \\ 0 &\leq \lambda_1, \quad 0 \leq \lambda_2, \quad 0 \leq \lambda_3. \end{aligned}$$

(2 marks) Any solutions to these conditions is maximizer of the Lagrangian, because the Lagrangian is a sum of concave functions by the arguments in (a) and because Slater condition holds.

(c) (20 marks) Suppose the restrictions on a found in part (a) hold. Solve the constrained optimization problem in terms of a .

Solutions: Consider the various supports:

- (2 marks) All three multipliers cannot be equal to 0, since FOC would be violated given the objective is increasing in both variables.
- (2 marks) All three multipliers can be strictly positive only if $a = \sqrt{3}/2$. If so, the three complementary slackness conditions would be met by setting $x(a) = 0$ and $y(a) = a$. But this would violate Slater condition.
- (7 marks) If $\lambda_2(a) = 0$ and $\lambda_1(a), \lambda_3(a) > 0$, then by complementary slackness $y(a) = a$ and $x(a) = (3/4 - a^2)^{1/2} > 0$ for $a < \sqrt{3}/2$. Further, FOC would then imply that

$$\begin{aligned} \lambda_1(a) &= \frac{a}{2(3/4 - a^2)^{1/2}} > 0 \\ \lambda_3(a) &= \frac{a^2}{(3/4 - a^2)^{1/2}} - 1 \geq 0 \Leftrightarrow a^4 + a^2 \geq 3/4. \end{aligned}$$

So as multipliers are non-negative, this is a solution for $a \in [1/\sqrt{2}, \sqrt{3}/2)$.

- (2 marks) If $\lambda_3(a) = 0$ and $\lambda_1(a), \lambda_2(a) > 0$, then by complementary slackness $x(a) = 0$ and $y(a) = \sqrt{3}/2$. Further, FOC would then imply that $\lambda_1(a) = 1/\sqrt{3} > 0$ and $\lambda_2(a) = -a < 0$. This cannot be a solution.
- (7 marks) Finally, if $\lambda_3(a) = \lambda_2(a) = 0$ and $\lambda_1(a) > 0$, then FOC and the constraints would imply that

$$\begin{aligned} y(a) &= \sqrt{\frac{3}{4 + 4a^2}} \geq a \Leftrightarrow a \leq 1/\sqrt{2}, \\ x(a) &= \sqrt{\frac{3a^2}{4 + 4a^2}} \geq 0, \\ \lambda_1(a) &= \frac{1}{2y(a)} \geq 0 \Leftrightarrow a \leq 1/\sqrt{2}. \end{aligned}$$

So as constraints need to be satisfied, this is a solution for $a \in (0, 1/\sqrt{2}]$.

(d) (6 marks) Suppose the restrictions on a found in part (a) hold. Use the Envelope Theorem to express the slope of value function with respect to the parameter a in terms of the solution $(x(a), y(a), \lambda(a))$ found in part (c).

Solutions: (6 marks) The Envelope Theorem states that if $(x(a), y(a), \lambda(a))$ are differentiable functions and the constraint qualification holds, then

$$\frac{dv(a)}{da} = \frac{\partial L(x, y, \lambda|a)}{\partial a} \Big|_{(x, y, \lambda) = (x(a), y(a), \lambda(a))} = x(a) - \lambda_3(a).$$

4. (20 Marks) Consider the correspondence $\mathbf{h} : [0, 2]^2 \rightrightarrows [0, 2]^2$, where $\mathbf{h}(\mathbf{x}) = (h_1(\mathbf{x}), h_2(\mathbf{x}))$ and

$$h_1(\mathbf{x}) = \begin{cases} 2 & \text{if } x_2 \in (1/2, 3/2) \\ [0, 2] & \text{if } x_2 \in \{1/2, 3/2\} \\ 0 & \text{if } x_2 \in [0, 1/2) \cup (3/2, 2] \end{cases} \quad \text{and} \quad h_2(\mathbf{x}) = \begin{cases} [x_1^2/2, x_1^{0.5}/2] & \text{if } x_1 \in [0, 1] \\ [x_1^{0.5}/2, x_1^2/2] & \text{if } x_1 \in [1, 2] \end{cases}.$$

Argue why *Kakutani fixed point theorem* applies. Then, find all the *fixed points* of this correspondence.

Solutions: (3 marks) Both correspondences have closed graph and are therefore UHC. Moreover, both are obviously convex-valued and non-empty. Since the image of $\mathbf{h}$ belongs to $[0, 2]^2$ which is also the domain, the Fixed Point Theorem applies. Thus, there will be at least one fixed point on $[0, 2]^2$.

(2 marks) First, consider $(0, 0)$. Such a point is a fixed point of $\mathbf{h}$ since $(0, 0) = \mathbf{h}(0, 0)$. (3 marks) Next, consider $(1, 1/2)$. Such a point is a fixed point of $\mathbf{h}$ since

$$(1, 1/2) \in [0, 2] \times \{1/2\} = \mathbf{h}(1, 1/2).$$

(5 marks) Next consider $(x_1, 3/2)$ for $x_1 \in [\sqrt{3}, 2]$. Such points are fixed points of $\mathbf{h}$ since

$$(x_1, 3/2) \in [0, 2] \times [x_1^{0.5}/2, x_1^2/2] = \mathbf{h}(x_1, 3/2) \quad \text{for } x_1 \in [\sqrt{3}, 2].$$

(5 marks) Finally, consider $(2, x_2)$ for $x_2 \in [1/\sqrt{2}, 3/2]$. Such points are fixed points of $\mathbf{h}$ since

$$(2, x_2) \in \{2\} \times [1/\sqrt{2}, 2] = \mathbf{h}(2, x_2) \quad \text{for } x_2 \in [1/\sqrt{2}, 3/2].$$

(2 marks) There are no fixed points in which $x_2 \in (0, 1/2) \cup (3/2, 2]$ since $h_1(\mathbf{x}) = 0$ and $h_2(0, x_2) = 0$. There are no fixed points in which $x_2 \in (1/2, 1/\sqrt{2})$ since $h_1(\mathbf{x}) = 2$ and $h_2(2, x_2) = [1/\sqrt{2}, 2]$.